



40mm × 40mm × 5mm · 316L Stainless Steel

COSMO

Connected Intelligence Architecture

Hardware Manual v1.0 · 100-Unit Production Run

Inventor: Vadik Marmeladov – COSMO® CIA · LOT Systems

```
> COSMO CIA v1.0 BOOT SEQUENCE
> WiFi: connecting .....
> LOT API: authenticated .....
> BME688 IAQ: calibrating ....
> E-paper: initialized .....
> Notifications: polling .....
>
> READY – Connected to lot-systems.com
```

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[PHYSICAL SPECIFICATION :: 316L STAINLESS STEEL]

SIDE A –

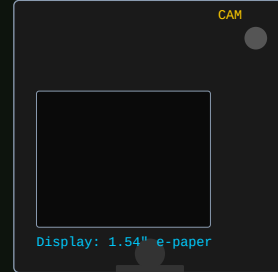


Mirror polish $Ra \leq 0.1\mu m$

No features / No text

40mm × 40mm · 2.8mm thick

SIDE B – ACTIVE FACE



CAM

Display: 1.54" e-paper

USB-C (flash)

>> DIMENSIONS & MATERIALS

Footprint	40.0 × 40.0 mm
Thickness	5.0 mm total (2.8mm Side A + 0.2mm gasket + 1.7mm Side B + 0.3mm PCB sta
Weight	~38 g (estimated)
Body material	316L Stainless Steel (2 CNC machined parts)
Side A finish	Mirror polish – $Ra \leq 0.1\mu m$ – ASTM A967 passivation
Side B finish	Satin brushed – $Ra 0.8\mu m$ – ASTM A967 passivation
Fasteners	4× M1.6 × 4mm 316L SS recessed screws
Seal	0.3mm silicone gasket – IP52 dust/splash resistant

>> SIDE B FEATURE POSITIONS

E-paper display window	28.0 × 38.0 mm	Center face, 6mm from top edge
Camera aperture	Ø3.5 mm	Top-right corner – 35mm from left, 3.5
Copy button	Ø7.0 mm recess, Ø2.5 mm through	Bottom-center – 37mm from top
Charge LED	Ø1.2 mm light pipe	Bottom-left – 3mm from left, 37mm from
USB-C slot	9.5 × 4.0 mm	Bottom edge center (flash/debug only)

[BILL OF MATERIALS :: 100-UNIT RUN]

>> KEY COMPONENTS

Ref	Part	Mfr	Source	\$/unit	Notes
U1	ESP32-S3-MINI-1-N8	Espressif	LCSC C2913202	\$3.50	MCU: Xtensa LX7 240MHz, WiFi, BLE 5
DISP1	GDEH0154D67	Good Display	Waveshare	\$4.20	1.54" e-paper 200×200 BW SPI
CAM1	OV2640 Module	OmniVision	LCSC / AliExpress	\$2.80	2MP DVP, 24×24mm, flush lens
U3	BME688	Bosch Sensortec	LCSC C2682739	\$7.55	T/H/P/Gas AI IAQ sensor (BSEC2)
L1	WR135-30003	Würth Elektronik	Mouser 710-760308103	\$2.80	Qi RX coil 30mm, 0.6mm thick
U4	STWLC68	STMicroelectronics	Mouser 511-STWLC68JR	\$2.20	Qi 5W WPC 1.2.4 receiver IC
BAT1	LP303030	Generic LiPo	LCSC bulk	\$3.20	200mAh 3.7V 30×30×3mm
U5	BQ21040DBVR	Texas Instruments	Mouser 595-BQ21040DBVR	\$0.95	LiPo charger 500mA, SOT-23-5
U6	MAX17048G+T	Analog Devices	Mouser 700-MAX17048G+T	\$1.60	Fuel gauge I2C 1% accuracy
SW1	TS-1187A-B	XKB	LCSC C318937	\$0.08	6mm tactile, 160gf, 0.5mm travel
—	Passives bundle	Various	LCSC	\$0.50	Caps, resistors, ferrites, ESD

>> COST SUMMARY (100 UNITS)

PCBWay PCBA (110 boards, turnkey)	\$12.50/unit	\$1,375
CNC Enclosure, 316L SS, 110 sets	\$22.60/unit	\$2,486
Electronic components (BOM above)	\$24.50/unit	\$2,695
Packaging (box, foam, cable, card)	\$2.60/unit	\$286
Prototype run + validation (Phase 1)	—	\$350
Logistics + duties + contingency 15%	—	\$1,089
TOTAL — 100 units shipped	~\$8,281	(~\$83/unit landed)

[PCBWAY MANUFACTURING GUIDE :: PCB + PCBA + CNC]

>> PCB SPECIFICATION

Board size	36 × 36 mm
Layer count	4 layers
Board thickness	1.0 mm
Surface finish	ENIG (Electroless Nickel Immersion Gold)
Copper weight	1 oz outer / 0.5 oz inner
Solder mask	Black (both sides)
Silkscreen	White (Side B component side)
Via treatment	Tented
Quantity to order	110 boards (100 production + 10 spare)
Lead time	3-5 business days

>> PCBA ASSEMBLY SERVICE

Assembly type	Double-sided SMT (reflow oven, lead-free RoHS)
Sourcing mode	Turnkey (PCBWay sources from LCSC)
Inspection	AOI all boards + X-ray for STWLC68 (QFN-28)
Critical no-substitute	U1 ESP32-S3, U3 BME688, U4 STWLC68, U6 MAX17048
Hand-solder after PCBA	FPC connector, OV2640, LiPo battery, SW1 button
Files required	Gerbers ZIP + BOM XLSX + CPL CSV

>> CNC ENCLOSURE ORDER (316L SS)

Side A – polished face	40×40×2.8mm, mirror Ra≤0.1μm, M1.6 inserts, STEP file
Side B – active face	40×40×2.5mm, satin Ra0.8μm, all cutouts, STEP file
Quantity	110 sets (220 individual shells)
Post-processing	Passivation ASTM A967 required for both shells
Tolerance	±0.05mm cavity / ±0.1mm outer dimensions
Lead time	10-12 business days

>> ORDER TIMELINE (PARALLEL)

[FIRMWARE & LOT API CONNECTOR :: ESP32-S3]

>> FIRMWARE QUICK REF

Framework	Arduino ESP32 Core 3.x
MCU	ESP32-S3-MINI-1-N8 @ 240MHz
Flash scheme	OTA dual-bank (app0/app1)
WiFi setup	WiFiManager captive portal
Notif poll	Every 30 seconds
Sensor upload	Every 5 minutes
Heartbeat	Every 10 minutes
Light sleep	25s between polls (~2mA)
Deep sleep	After 30min idle (~120µA)
Battery life	7-10 days normal use
Factory reset	Hold button 10s at boot

>> API ENDPOINTS

>> COPY BUTTON GESTURES

```
>> E-PAPER DISPLAY LAYOUT (200x200 px)
```

```

#####
■  LOT                BAT: 87%    ■  ← Status bar (20px)
#####
■
■  Coffee time!          ■  ← Notification (18
■
■  11:34 AM             ■  ← Timestamp
■  via: memory_engine    ■  ← Source
■
#####
■  [0] Copy            CHG        ■  ← Footer (20px)
#####

```

[PROJECT ROADMAP :: 100-UNIT PRODUCTION RUN]

Phase 0	Design Lock	✓ Initiated
Weeks 1-2	PCB schematic, layout, STEP files, BOM finalized	
Phase 1	PCB Prototyping	■ Pending
Weeks 3-5	10 prototype boards – functional test + field validation	
Phase 2	Firmware Alpha	■ Pending
Weeks 3-6	WiFi, display, BME688, API, button, camera, OTA	
Phase 3	Prototype Validation	■ Pending
Weeks 5-7	5 internal testers – battery life, WiFi resilience	
Phase 4	Production Order	■ Pending
Weeks 7-10	Place PCBWay PCBA (110) + CNC (110 sets) orders	
Phase 5	Assembly & QA	■ Pending
Weeks 10-12	FPC, camera, battery, enclosure – 100% test every board	
Phase 6	Software Integration	✓ API built
Weeks 8-12	Device API, Settings page, Memory Engine notifs	
Phase 7	Distribution	■ Pending
Weeks 12-14	100 units shipped – Usership members first	

>> KEY RISKS

BME688 stock

Med

Order +20 units; alt: BME280+SGP40

CNC quality

Low

Prototype enclosure sample in Phase 1

Assembly yield

Low

10 spare boards as buffer (110 ordered)

OTA infra delay

Med

Non-blocking for v1.0 ship date

■	COSMO® CIA – Connected Intelligence	■
■	Architecture	■
■		■
■	40mm × 40mm × 5mm	■
■	316L Stainless Steel · E-paper · WiFi	■
■	BME688 AI Sensor · Qi Wireless · Camera	■
■		■
■	Copy button → lot-systems.com/log	■
■	Notifications ← Memory Engine AI	■
■	Sensor data → QOS Biofield Engine	■
■		■
■	lot-systems.com · brand.lot-systems.com	■

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